ECDLP index-calculus leads in the conversation export

Assessment date: 2026-10-12 (America/Los_Angeles). **Status:** source review and research decision; no new experiment or performance claim. **Source:** user-supplied private `conversations.json`, 154,481,197 bytes, SHA-256 `bf50fe6cbffc7a84e76d3c7b61c5a49d534a12f62a6b86a6c3b3ec50d9685528`. The export is not copied into this repository. Conversation UUIDs below identify discussions, not independent evidence. This review indexed 974 conversations, screened 67 with a relevant human prompt and 60 by title, then read the selected IC discussions. It does not validate every assertion in the export. Repository evidence is checked against `main` through merged PR #1722.

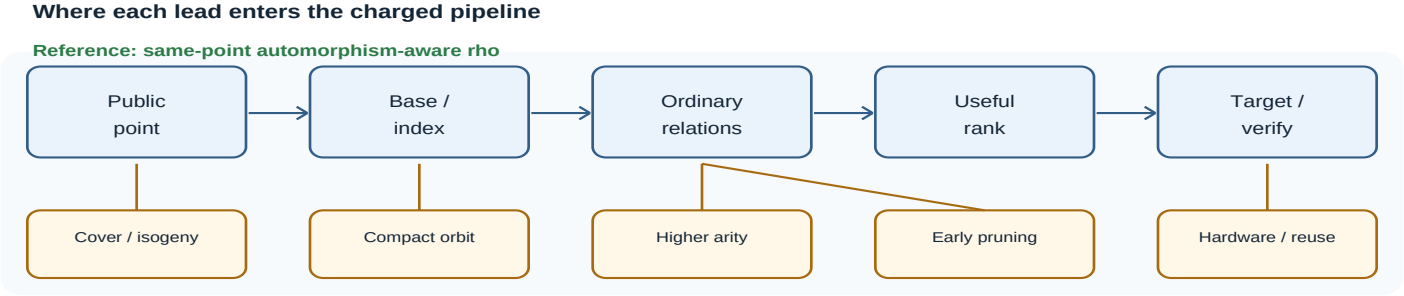
Question and comparison rule

Which ideas in the export most directly address relation supply, useful rank, decomposition cost, or transfer to an easier DLP? The reference is the same-point, automorphism-aware rho in the boundary ledger. Keep reusable setup, full cold work, and one-target online work separate. The common operation-normalized S remains unmeasured for the current candidates; native S3 probes cannot be divided by rho walk steps as if they were the same unit. A controlled `vs_rho` claim needs the ledger's replay, exclusive-phase, calibration, and host-isolation receipts.

What the current record establishes

Regime and source	Recorded observation	Meaning for this assessment
Disjoint cold compact-orbit panel	IC/rho complete-child-CPU median 2.810 at n37/K7/one target, 13.378 at n41/K85/one target, and 22.675 at n53/K220/one target; all six preregistered cells exceeded parity.	The measured cold policies have a setup and rank-cost deficit. The separate 1,024-target cells are amortized evidence.
n83 a=1/K600 strong-rho study	Five repeats of one Q: IC/strong online geometric mean 1.382 [1.036, 1.882]; fresh-process ratio 218.035 [172.358, 285.574], excluding original base construction.	The old 8.2 rho/IC observation used a legacy packed rho backend and is superseded as a reference. This study is diagnostic on one target and host.
n83 a=1/K600 tuned-rho follow-up	Five new targets after one 1,149 s IC preparation: IC/selected-rho online geometric ratio 0.431 [0.138, 1.776]; arithmetic ratio 1.543. IC was faster on four targets.	The interval includes parity and the two estimands give opposite orderings. One shared setup is not five cold single-target runs.
n53 fixed-base rank restart	Median rank-probe reduction 5.189%, paired complete-cold ratio 1.020; preregistered 15% gate failed.	Avoid assuming that reducing rare long rank searches lowers total cost.
n53 pair-swap quotient	Index S3 construction calls 2,565,200 to 1,282,710; six-seed paired median complete-cold ratio 0.969 and target-online ratio 1.095.	Exact index construction saving; no measured end-to-end gain in this panel.
n83 a=0/81-bit high-arity Boolean gate	Planted replay passed; ordinary public PDP reached the 120 s cap with `UNKNOWN`.	Circuit correctness established; ordinary relation yield unresolved. This is a different curve and subgroup from the a=1/K600 rows.
`cryptanalysis:experiments/ecc2k130-ic-goal-20260929/n131_w15_fixedseed_s3_join_20261010/RESULT.md` (local unmerged commit `ef2fab306e87`)	One constructed four-atom hit; both admissible published-point lifts had zero hits in those two fixed pair domains.	Exact component filter, with no whole-family coverage or rank measurement; this local record is not accepted evidence.

The diagram lead-map.svg shows where each proposed mechanism must enter the charged pipeline. Its editable source is lead-map.mmd. The a=1/K600 strong study independently replayed its completed scalar answers and retained its full phase/resource receipts; the full rank/factor-log transcript, common calibrated S, and independent-host/L2 qualification remain open. The boundary ledger has not promoted this study to controlled `vs_rho`.



Ranked leads and cheapest falsifiers

1. **Compact-orbit factor bases plus cheaper rank and target decomposition.** This is the most evidenced executable IC path in the export (conversations `877d18ed-ed3d-4701-9526-caafccab2de0`, `a3bb6356-5815-4a0b-8761-c4f4f4cb95d1`, `f16172d4-6740-4fce-93a8-997dda6936db`; subfield-curve paper). Frobenius collapses columns, but the cold panels show that base/index/rank preparation still dominates fresh one-target cost. The latest a=1/K600 fresh-target panel gives mixed prepared-online results against tuned strong rho, strengthening the case for reducing preparation without promoting an online winner. The immediate falsifier is a frozen, fixed-useful-column comparison on new disjoint points: count natural relation yield, novel rank per accepted row, complete base/index/rank and target phases, bytes, and matched optimized rho. First isolate the hot second S3 root call under an identical rank trace; the pair-swap result indicates why an index-only saving can miss the target path. Class: **engineering candidate**, with an open possibility of a new relation/rank regime only if the measured ratio to the fixed boundary changes.

2. **Target-coupled higher-arity joins and alternative factor-base geometry.** Conversations `877d18ed-ed3d-4701-9526-caafccab2de0`, `d57e1315-4f00-4f55-92a4-97d9b9cd66f6`, and `ee67891a-e959-4dbb-9dd1-b47f2b4dfe60` propose more summands, shifted/Frobenius domains, and exact pair joins. The n131 W15 result proves a fixed-family filter, while the public-point misses and n83 bounded PDP show why supply and solve cost must be measured jointly. The next bounded gate is the existing N41 target-coupled seed-family elimination, with ordinary targets, constructed correctness controls, complete candidate-family counts, useful-rank checks, memory, and charged work; advance toward N131 only with a validated scaling law and a native Rust execution path. Class: **proposed decomposition route**, not an observed complete-method improvement.

3. **Early-prunable algebraic or SAT/F4/F5 systems.** Conversations `370c1202-ba41-4a45-ba9c-c41a7c3bf8f0`, `6108ece5-7c36-4f1b-94f7-2b837d3318b3`, `d57e1315-4f00-4f55-92a4-97d9b9cd66f6`, and `59fceced-c301-4865-a848-0e04b0a047fb` focus on first-fall degree, solver structure, syzygies, and symmetry (see the WDSat paper and last-fall-degree analysis). A smaller formula or fewer SAT decisions matters only if it changes ordinary-query cost under the same complete factor-base domain, or improves a full charged run. The falsifier is a frozen ordinary-query panel with valid SAT, UNSAT, timeout, and producer-failure distinctions, matched formula domains, exact group replay, and a measured growth curve across sizes. The n83 120 s `UNKNOWN` is censored, not evidence of no relation. Class: **unverified mechanism with possible exponent relevance**; the export's categorical claims about first-fall degree should not be treated as a universal theorem.

4. **Prime-field covers, Jacobian representations, and isogeny transport.** Conversations `49ed070e-eeaa-46f4-bafe-3af132e055ae`, `ce127e8d-13d1-4f58-ac88-15e038b856b9`, and `c68795ee-9d88-4ba7-ae79-4a2d483729dd` raise the highest-upside structural question for prime fields. Their existence claims do not yet supply an executable target map, preserved prime-order subgroup, cheaper destination DLP, or charged setup/transfer/recovery. A prime extension degree excludes a *proper intermediate subfield* in the particular Joux-Vitse tower route; it does not exclude every cover or alternative representation. The cheapest falsifier is one exact curve/subgroup and explicit map with independent group-law replay, then a destination relation-yield/rank test with map costs and matched rho. Class: **speculative transfer hypothesis**. The endomorphism rules govern this distinction.

5. **Multiple targets and hardware acceleration.** Conversations `f16172d4-6740-4fce-93a8-997dda6936db`, `9989b36c-cace-48bc-b5ef-92f946a543f6`, and `33371c38-2625-4bfe-8f59-08cf777b3faa` discuss cached algebra and GPU/TPU execution. These can lower amortized or primitive cost, but require setup divided by the actual number of reused targets and a separately reported one-target result. They do not themselves establish a better mathematical relation mechanism. Class: **secondary amortization and engineering**.

Decision and unresolved work

First run the fixed-useful-base S3/rank cost experiment, then the N41 target-coupled family-elimination gate. Keep early-pruning solvers as a parallel mathematical design track and prime-field transfers as an explicit-map track. No experiment was run for this assessment. The next experiment requires a versioned protocol with frozen inputs, a same-point rho reference, success and stop conditions, and complete phase accounting. Existing private conversation text is context; only committed receipts can promote a finding.

Checked canonical graphs: `docs/index-calculus-scoreboard.html` and `docs/ic/progress-timeline.json`. They remain unchanged because this review adds no verified numerical result or correction to a canonical measurement.

To rebuild the PDF, run `python3 render_report.py` in this directory with ReportLab 4.5.1. The Python script only lays out this recorded assessment and diagram; it performs no cryptographic computation.